

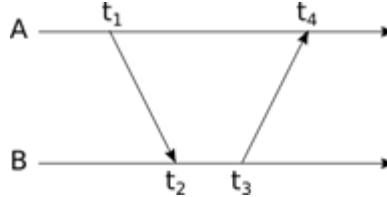
CSC0056 Data Communication, Homework 5

Department of Computer Science and Information Engineering
National Taiwan Normal University

Dec. 10, 2024

Nine questions. 100 points total.

1. (10 points) Consider the following timelines that illustrate the message exchanges for NTP clock synchronization. Assume that A is a local embedded system running a NTP client and that B is a cloud workstation running a NTP server. Assume the four UNIX timestamps the embedded system may get are $t_1=1741688100$, $t_2=1742789000$, $t_3=1742789005$, $t_4=1741688115$. Compute θ the clock offset between the two clocks ¹.

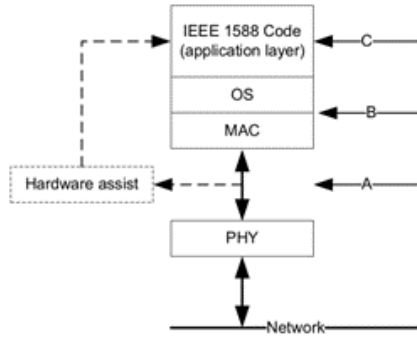


2. (10 points) According to the arbitration process in the CAN protocol, using the wired-AND implementation, what would be the correct sequence of three-round transmission of the following three frames if they all arrived at the CAN bus in the current round?

F1: 011100110011
F2: 010110110010
F3: 010110110011

3. (10 points) In the PTP protocol, consider the timestamp generation model as follows (copied from Figure 18 in the IEEE 1588-2019 standard). To get a better time synchronization result, is it better to take timestamp at A or at C ? Explain.

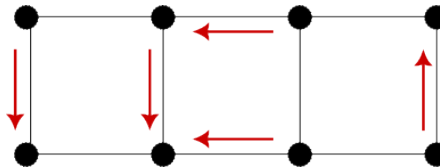
¹For your information, the UNIX timestamp represents “the number of seconds that have elapsed since January 1, 1970 (midnight UTC/GMT), not counting leap seconds”; see <https://www.epochconverter.com/>.



4. (15 points) Briefly explain why the PTP use of transparent clocks may improve the precision of one-way delay measurement.
5. (10 points) Give an example pattern of six errors that will escape the detection by the horizontal and vertical parity checks.
6. (10 points) In CRC, let $g(D) = D^4 + D^2 + D + 1$ and let $s(D) = D^3 + D$. Compute $x(D)$ the frame to be transmitted.
7. (10 points) Regarding the I2C protocol, is the following statement true or false?

Statement: If there are multiple clocks attached to the I2C clock line, the clock with shortest HIGH period will become the HIGH period for the synchronized clocks.

8. (15 points) For the following wireless network topology, given those five one-way concurrent transmissions, determine n_p the number of primary interferences and n_s the number of secondary interferences. Count all nodes involving in a particular interference as one interference.



9. (10 points) What is the Hamming weight for the set of code words shown on Page 8 of the slides for Week 14?